

## Problem Set 2

Solutions by Harsh Bhalala

**Problem 1. Asymptotic Relationships**

(a)

| $f(n)$                               | $g(n)$                               | Relationship?     |
|--------------------------------------|--------------------------------------|-------------------|
| $6^n$                                | $n^{1200}$                           | $f \in \omega(g)$ |
| $(\log n)^6$                         | $\sqrt[1200]{n}$                     | $f \in o(g)$      |
| $\log_6(n^{1200})$                   | $(\log_{1200} n)^6$                  | $f \in o(g)$      |
| $\log(n!)$                           | $n \log n$                           | $f \in \Theta(g)$ |
| $2^{\lceil \log_2 n \rceil}$         | $3^{\lceil \log_3 n \rceil}$         | $f \in \Theta(g)$ |
| $2^{\lceil \log_2 n \rceil} - n + 1$ | $3^{\lceil \log_3 n \rceil} - n + 1$ | None.             |
| $n$                                  | $n^{\cos \log n}$                    | None.             |

(b)

| $f(n)$           | $g(n)$                         | Relationship?     | Justification                                            |
|------------------|--------------------------------|-------------------|----------------------------------------------------------|
| $2^{(\log n)^2}$ | $2^{(\lceil \log n \rceil)^2}$ | $f \in O(g)$      | $\lceil \log n \rceil \geq \log n$                       |
| $4^n$            | $(2n)!/(n!)^2$                 | $f \in \omega(g)$ | $\lim_{x \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$ |

**Problem 2. OEIS**

(a)

$$x_3 = 5, x_4 = 12, x_5 = 29, x_6 = 70$$

(b)(c) Found formula:

$$a(n) = \frac{(1 + \sqrt{2})^n - (1 - \sqrt{2})^n}{2\sqrt{2}}$$

(d)

*Proof by strong induction.* Induction hypothesis which says, sequence that begins with  $x_0 = 0, x_1 = 1, x_2 = 2$  and continues with  $x_n = 5x_{n-2} + 2x_{n-3}$  for  $n \geq 3$ , nth term can also be calculated via  $P(n) = \frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{2\sqrt{2}}$  for  $x \geq 0$

Base cases:  $P(0) = 0$ 

$$P(1) = \frac{(1 + \sqrt{2})^1 - (1 - \sqrt{2})^1}{2\sqrt{2}} = 1$$

$$P(2) = \frac{(1 + \sqrt{2})^2 - (1 - \sqrt{2})^2}{2\sqrt{2}} = 2$$

Inductive step: For strong induction, assume  $P(n-1), P(n-2)$  is true for  $n \geq 2$ , and we want to show that  $P(n+1)$  is also true. By the definition of the given sequence, we know that

$$\begin{aligned}
x_{n+1} &= 5x_{n-1} + 2x_{n-2} \\
&= 5 \left( \frac{(1+\sqrt{2})^{n-1} - (1-\sqrt{2})^{n-1}}{2\sqrt{2}} \right) + 2 \left( \frac{(1+\sqrt{2})^{n-2} - (1-\sqrt{2})^{n-2}}{2\sqrt{2}} \right) \\
&= \frac{1}{2\sqrt{2}} \left[ \left( 5(1+\sqrt{2})^{n-1} + 2(1+\sqrt{2})^{n-2} \right) - \left( 5(1-\sqrt{2})^{n-1} + 2(1-\sqrt{2})^{n-2} \right) \right] \\
&= \frac{1}{2\sqrt{2}} \left[ (1+\sqrt{2})^{n-2} [7+5\sqrt{2}] - (1-\sqrt{2})^{n-2} [7-5\sqrt{2}] \right] \\
&= \frac{1}{2\sqrt{2}} \left[ (1+\sqrt{2})^{n-2} [1+3\sqrt{2}+3(\sqrt{2})^2+(\sqrt{2})^3] - (1-\sqrt{2})^{n-2} [1-3\sqrt{2}+3(\sqrt{2})^2-(\sqrt{2})^3] \right] \\
&= \frac{1}{2\sqrt{2}} \left[ (1+\sqrt{2})^{n-2} [1+\sqrt{2}]^3 - (1-\sqrt{2})^{n-2} [1-\sqrt{2}]^3 \right] \\
&= \frac{(1+\sqrt{2})^{n+1} - (1-\sqrt{2})^{n+1}}{2\sqrt{2}}
\end{aligned}$$

Which implies  $P(n+1)$ , and by the strong induction hypothesis,  $P(n)$  holds for all  $n \geq 0$ .  $\square$

### Problem 3. Bounded Asymptotics

(a)

*Proof.* Function  $f$  is bounded from above therefore,  $\exists B \in \mathbb{R} (f(n) \leq B)$  and since  $f : \mathbb{N} \rightarrow \mathbb{R}^+ (B > 0)$ . And the function  $g$  is bounded away from zero, meaning  $\exists A \in \mathbb{R} (A > 0 \text{ and } g(n) \geq A)$ . Now let  $c$  be a constant real number such that  $c = \frac{B}{A}$  and  $n_0 = 1$ , for all  $n > n_0$ , then,

$$\begin{aligned}
g(n) &\geq A \quad (\text{multiplying both side by } c) \\
c \cdot g(n) &\geq A \cdot c \\
c \cdot g(n) &\geq B \geq f(n) \\
c \cdot g(n) &\geq f(n)
\end{aligned}$$

Therefore, by the definition of Big  $O$ , we can conclude that  $f(n) \in O(g(n))$ .  $\square$

(b) No it is not true

*Proof by counterexample.*  $f(n) = 1$ , which is bounded above by value 1, and  $g(n) = 1/n$ , which is bounded from below by value 1, but here clearly  $f(n) \notin O(g(n))$   $\square$

(c) Yea it is true that  $f \in O(g)$  because we can always take  $f(1)/g(1)$  as value of constant  $c$  and multiply with  $g(n)$  and since  $f$  is strictly decreasing function and  $g(n)$  is strictly increasing, condition  $f(n) \leq c \cdot g(n)$  will always hold.

No, it is not always true that  $f \in o(g)$ . For example,  $f(n) = 1 + (1/n)$  is strictly decreasing and  $g(n) = 2 - (1/n)$  is strictly increasing, but the limit of their ratio will be  $\frac{1}{2} \neq 0$ .

### Problem 4. Karatsuba's Algorithm

(a)

$$T(2) = 3T(1) + 10 * (2) = 3 + 20 = 23$$

$$T(3) = 3T(2) + 10 * (3) = 69 + 30 = 99$$

$$T(4) = 3T(2) + 10 * (4) = 69 + 40 = 109$$

(b)

$$\begin{aligned}
T(n) &= 3T\left(\frac{n}{2}\right) + 10n \\
&= 3 \left[ 3T\left(\frac{n}{2^2}\right) + 10\frac{n}{2} \right] + 10n \\
&= 3^2 T\left(\frac{n}{2^2}\right) + 3 \cdot 10\frac{n}{2} + 10n \\
&= 3^2 \left[ 3T\left(\frac{n}{2^3}\right) + 10\frac{n}{2^2} \right] + 3 \cdot 10\frac{n}{2} + 10n \\
&= 3^3 T\left(\frac{n}{2^3}\right) + 3^2 \cdot 10\frac{n}{2^2} + 3 \cdot 10\frac{n}{2} + 10n \\
&\vdots \\
&= 3^k T\left(\frac{n}{2^k}\right) + 3^{k-1} \cdot \frac{10n}{2^{k-1}} + 3^{k-2} \cdot \frac{10n}{2^{k-2}} + \cdots + \frac{10n}{2^0} \\
&= 3^{\log_2 n} T(1) + \left[ \left(\frac{3}{2}\right)^{k-1} + \left(\frac{3}{2}\right)^{k-2} + \cdots + \left(\frac{3}{2}\right)^0 \right] 10n \quad \left(\frac{n}{2^k} = 1, k = \log_2 n\right) \\
&= 3^{\log_2 n} + \left[ \frac{\left(\frac{3}{2}\right)^k - 1}{\frac{1}{2}} \right] 10n \\
&= 3^{\log_2 n} + \left[ \frac{\left(\frac{3}{2}\right)^{\log_2 n} - 1}{\frac{1}{2}} \right] 10n \\
&= 3^{\log_2 n} + \left[ \frac{3^{\log_2 n} - n}{\cancel{\mathcal{K}}} \right] 20\cancel{\mathcal{K}} \\
&= 21 \cdot 3^{\log_2 n} - 20n
\end{aligned}$$

(c) For Karatsuba's Algorithm  $f(n) = 10n$ ,  $a = 3$  and  $b = 2$ , so for master theorem value of  $\log_b^a = \log_2^3$  and  $f(n) \in O(n^{\log_2^3})$  because  $\log_2^3 > 1$ , which implies  $T(n) = \Theta(n^{\log_2^3})$  by following the case I of master theorem.

(d) Yes, it is. From part (c), we know Karatsuba's algorithm takes  $\Theta(n^{\log_2^3})$ , and the value of  $\log_2^3 \approx 1.58$ , meaning it takes  $\Theta(n^{1.58})$  steps, which is an improvement over the straightforward algorithm which takes  $\Theta(n^2)$  steps.